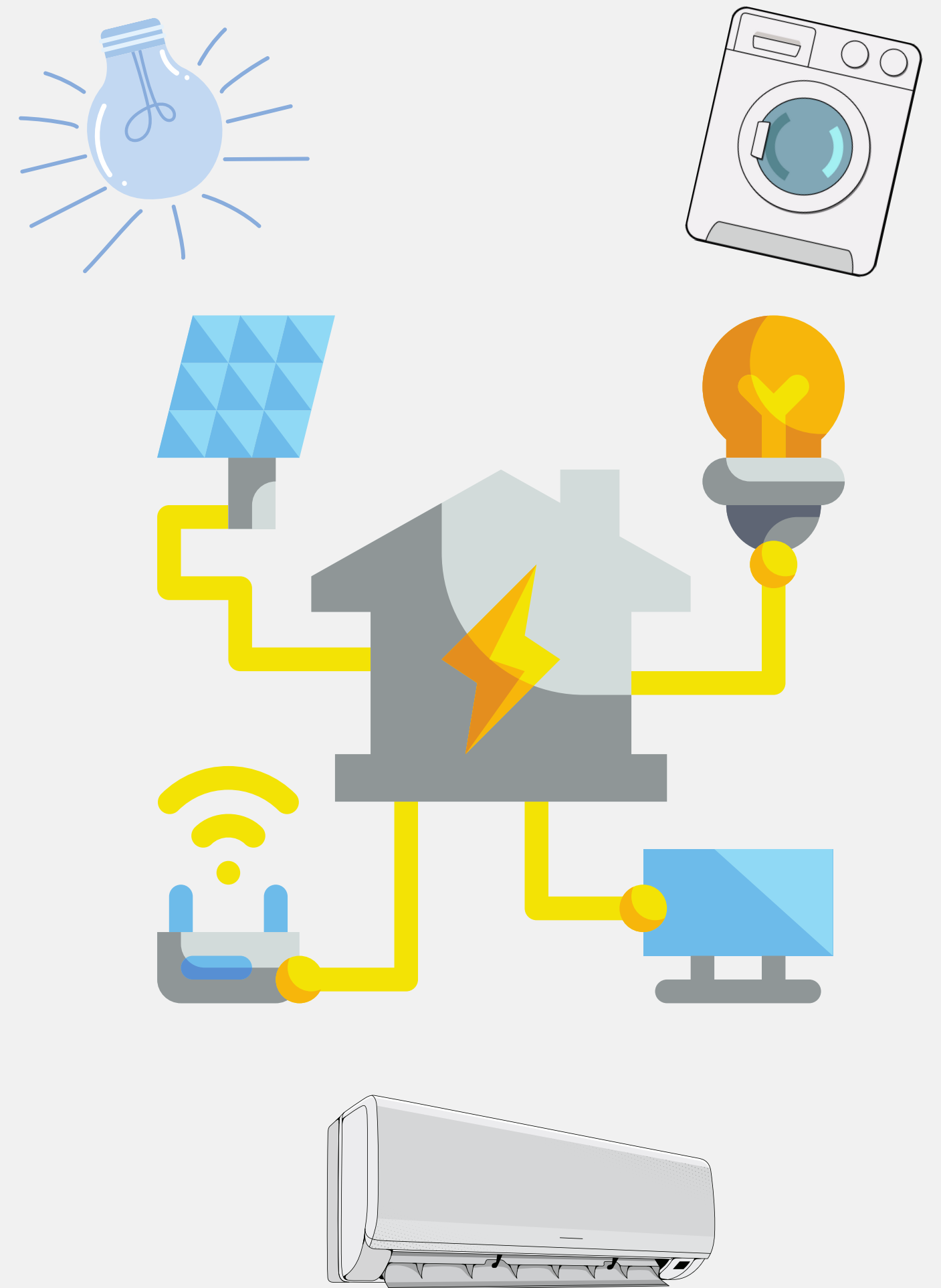
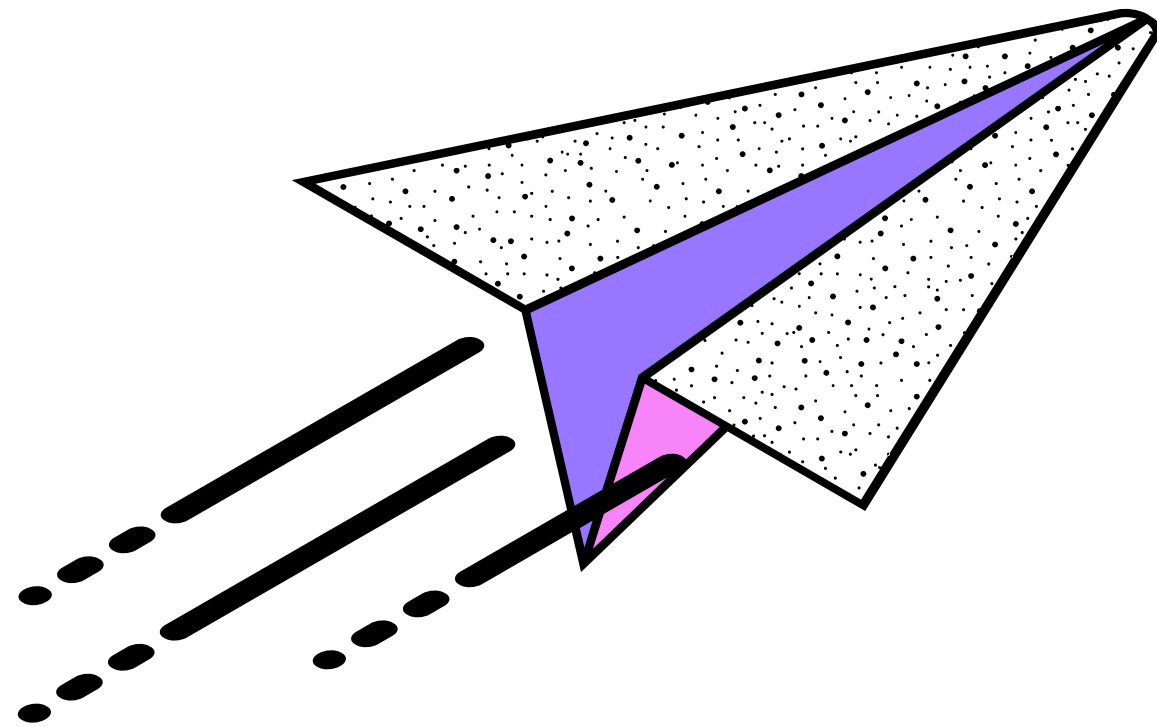


Development of a Nonintrusive Appliance Load Monitoring System(NIALM)



Outline



Introduction

Importance of Study

Problem Statement

Aims and Objective

Scope and Limitation of Study

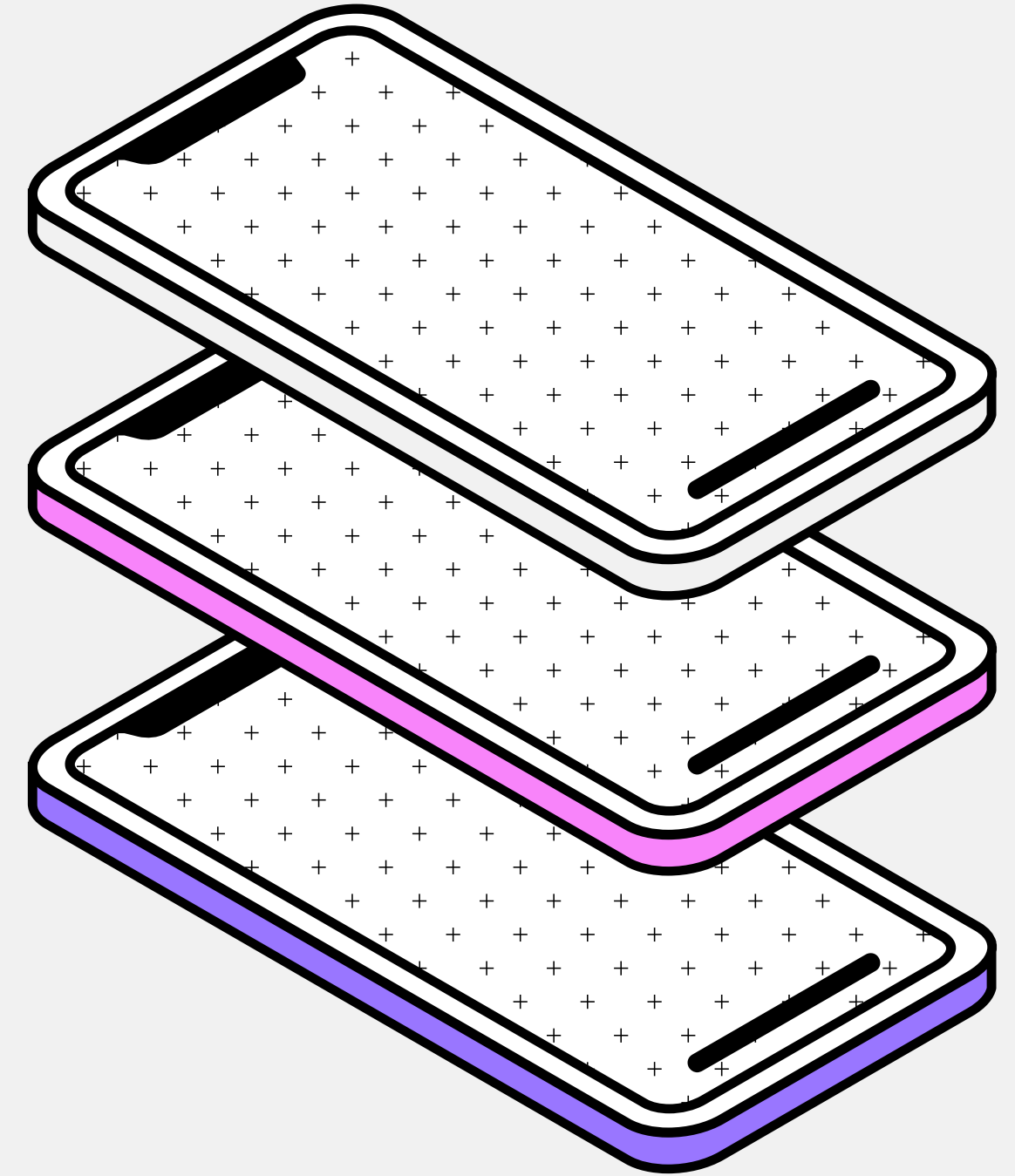
Methotology and Work Plan

Results and Discussion

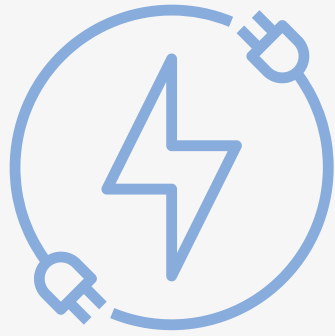
Recommendation for Future Improvements

Introduction about NIALM

- Monitors the usage of load or electricity of appliances
- No direct connection to the device or appliance
- Smart home concept with IoT implementation



Importance of Study



To supply information of electricity to consumers



To educate individuals about the practice of conserving energy



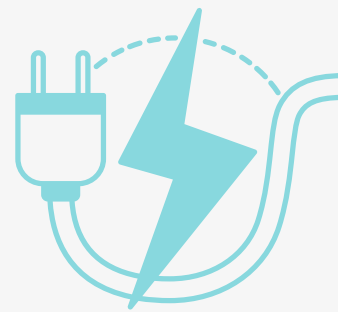
In the industrial sector, NIALM systems could recommend the most efficient designs of equipment.

Problem Statement



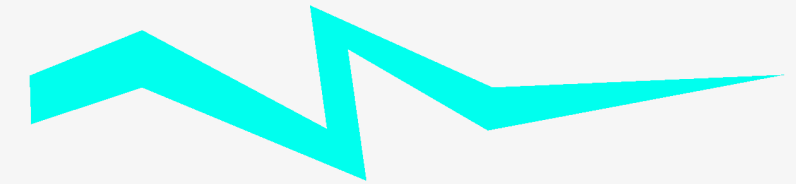
Increasing Cost

Cost is increasing by equipping every appliance with an individual meter.



Unstable Power Supply

Current that is running through the monitored appliances also go through the meters, which makes the power supply less stable.



Increase Energy Consumption

The greater the number of monitoring system elements, the higher the total energy consumption.

Aims and Objectives

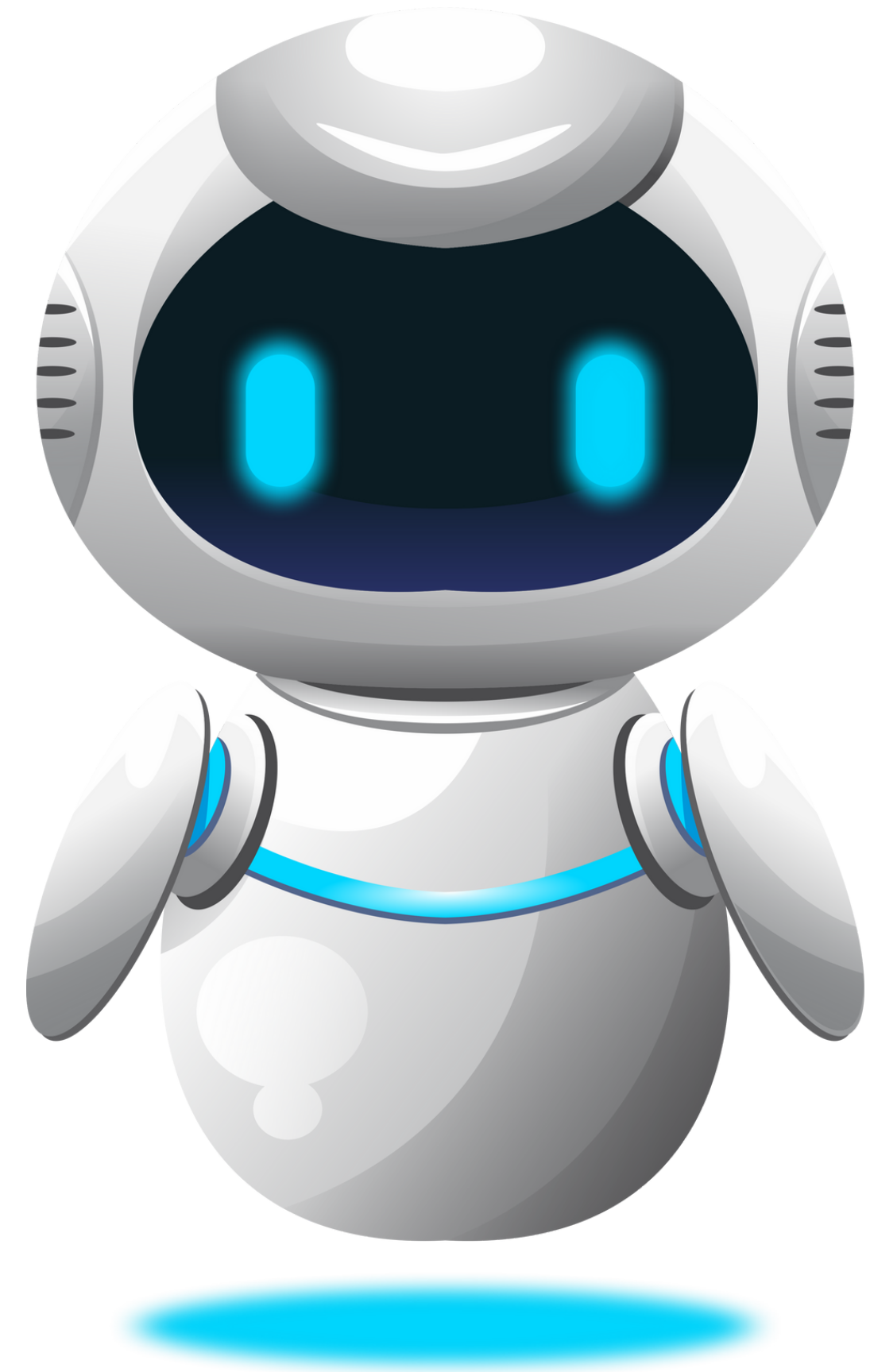
To develop a nonintrusive appliance load monitoring (NIALM) system for automatic home appliance classification and monitoring.



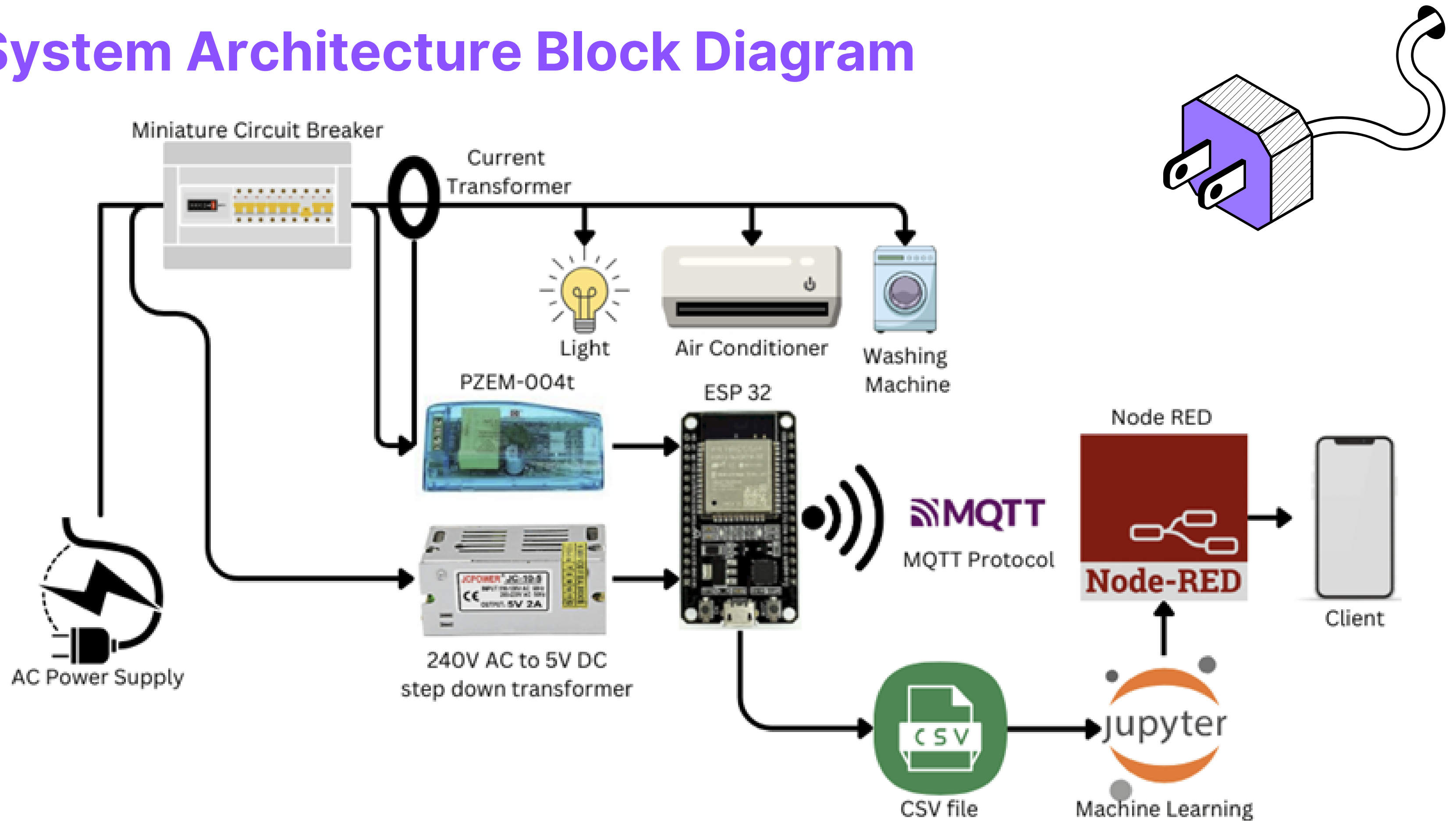
- To analyse NIALM technologies and a suitable design for the non-intrusive monitoring of the load produced by home appliances.
- To develop the proposed NIALM system, which can automatically identify household appliances and monitor the amount of energy that is used by appliances.
- To evaluate the performance of the developed system.

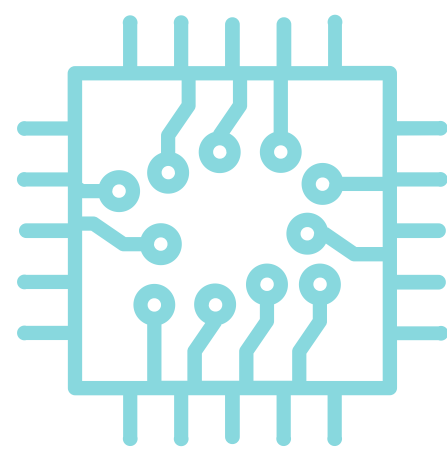
Scope and Limitations of NIALM

- Study and build a Non-Intrusive Appliance Load Monitoring system.
 - Study and try different algorithms and neural networks
 - Implement IoT in this system
-
- Set up communication protocol
 - Tuning machine learning model

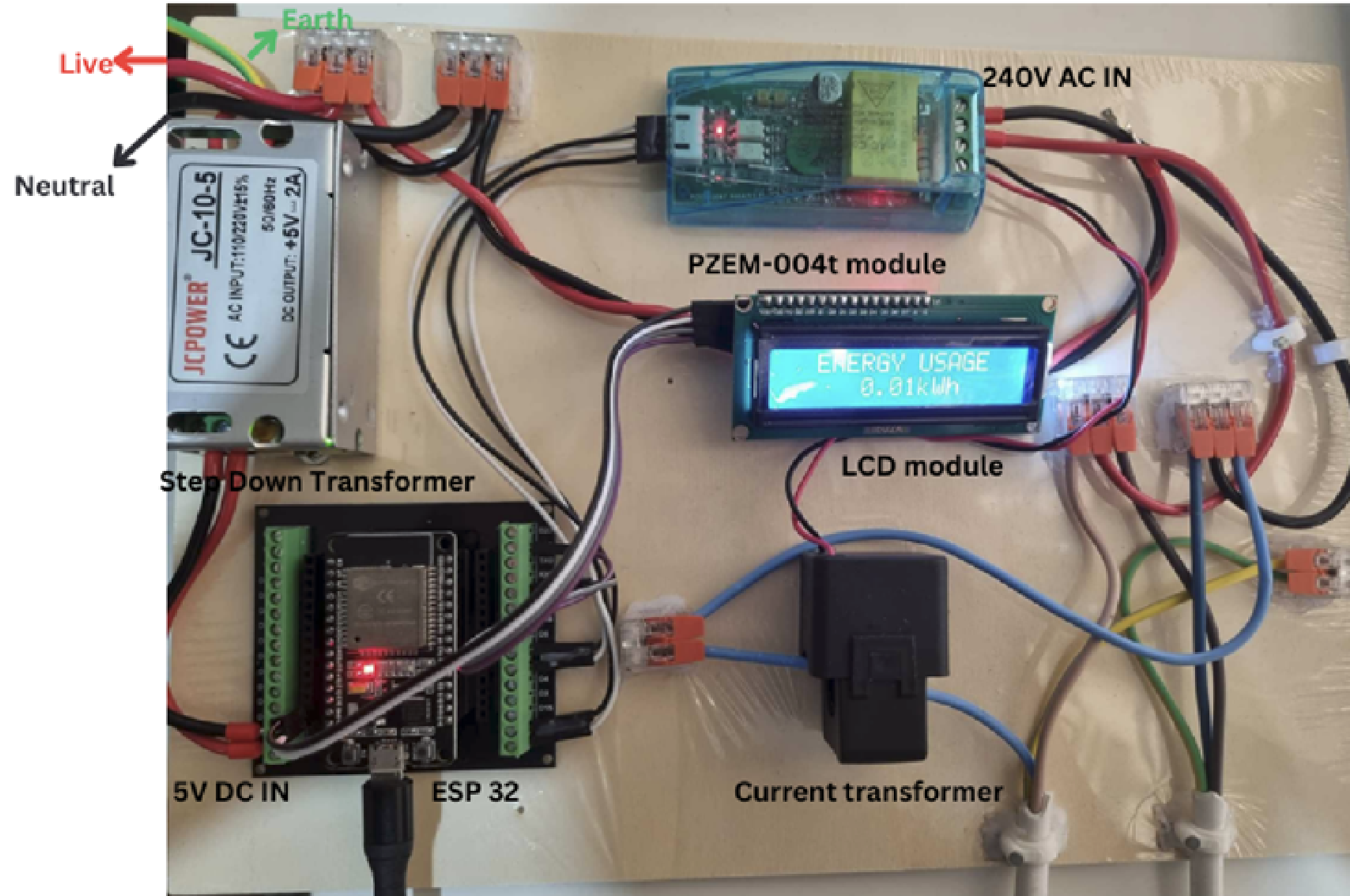
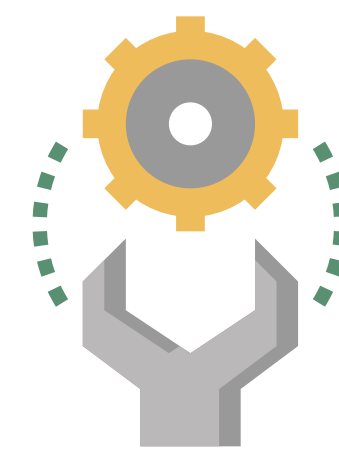


System Architecture Block Diagram



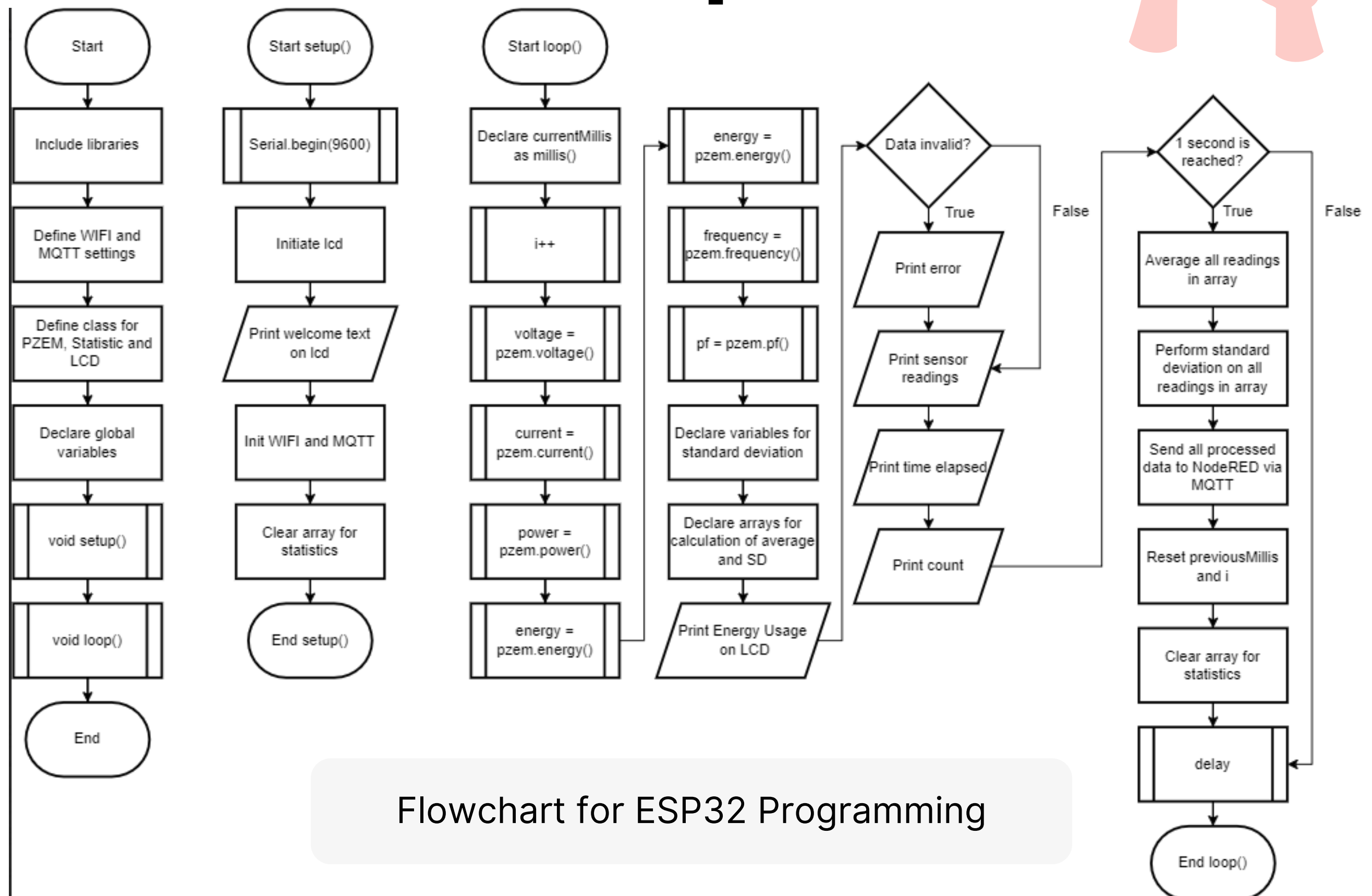


Hardware Design





Software Development



Flowchart for ESP32 Programming



Software Development



0 - OFF

1 - LIGHT

2 - AC

3 - WM WATER INFILL

4 - WM ROTATE SPIN

5 - WM CONTINUOUS SPIN

6 - LIGHT AC

7 - LIGHT WM WATER INFILL

8 - LIGHT WM ROTATE SPIN

9 - LIGHT WM CONTINUOUS SPIN

10 - AC WM WATER INFILL

11 - AC WM ROTATE SPIN

12 - AC WM CONTINUOUS SPIN

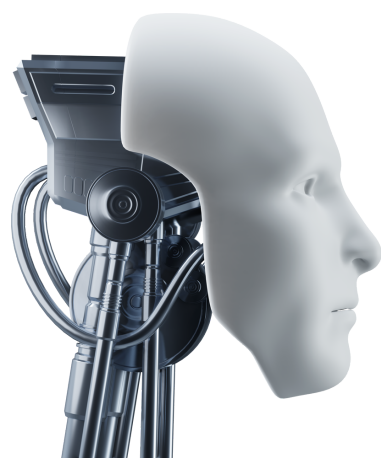
13 - LIGHT AC WM WATER INFILL

14 - LIGHT AC WM ROTATE SPIN

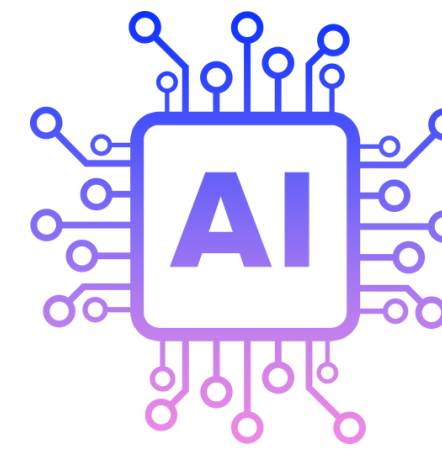
15 - LIGHT AC WM CONTINUOUS SPIN

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
DATE	TIME	TIME ELAPSED	COUNT	VOLTAGE	CURRENT	POWER	ENERGY	FREQUENCY	PF	SDVOLTAGE	SDCURRENT	SDPOWER	SDENERGY	SDFREQUENCY	SDPF
8/23/2023	3:26:51 PM	20741	10	244.1	0.1	11	6.562	50	0.45	0	0	0	0	0	0
8/23/2023	3:26:52 PM	21747	10	244.1	0.0991	10.91	6.562	49.97	0.45	0	0.0003	0.03	0	0.0458	0
8/23/2023	3:26:53 PM	22814	10	244.1	0.0997	10.9	6.562	49.93	0.45	0	0.0005	0	0	0.0458	0
8/23/2023	3:26:54 PM	23820	10	244.04	0.0994	10.9	6.562	49.93	0.45	0.049	0.0005	0	0	0.0458	0
8/23/2023	3:26:55 PM	24887	10	244.04	0.099	10.9	6.562	49.94	0.45	0.049	0	0	0	0.049	0
8/23/2023	3:26:56 PM	25834	9	244.1	0.099	10.9	6.562	49.9667	0.45	0	0	0	0	0.0471	0
8/23/2023	3:26:57 PM	26900	10	244.1	0.099	10.9	6.562	49.96	0.45	0	0	0	0	0.049	0
8/23/2023	3:26:58 PM	27907	10	244.01	0.099	10.9	6.562	49.96	0.45	0.03	0	0	0	0.049	0
8/23/2023	3:26:59 PM	28974	10	244.07	0.0997	10.97	6.562	49.93	0.45	0.0458	0.0005	0.0458	0	0.0458	0
8/23/2023	3:27:00 PM	29980	10	244.07	0.1	11	6.562	49.96	0.45	0.0458	0	0	0	0.049	0
8/23/2023	3:27:01 PM	31047	10	244	0.1	11	6.562	49.93	0.45	0	0	0	0	0.0458	0
8/23/2023	3:27:02 PM	31994	9	244.0889	0.0991	11	6.562	49.9	0.4589	0.0314	0.0003	0	0	0	0.0031
8/23/2023	3:27:03 PM	33060	10	244.1	0.0997	11	6.562	49.93	0.453	0	0.0005	0	0	0.0458	0.0046
8/23/2023	3:27:04 PM	34067	10	244.1	0.1	10.94	6.562	49.9	0.45	0	0	0.049	0	0	0
8/23/2023	3:27:05 PM	35134	10	244.1	0.1	10.94	6.562	49.94	0.45	0	0	0.049	0	0.049	0
8/23/2023	3:27:06 PM	36140	10	244.1	0.1	11	6.562	49.91	0.45	0	0	0	0	0.03	0
8/23/2023	3:27:07 PM	37207	10	244.1	0.1	11	6.562	49.9	0.45	0	0	0	0	0	0
8/23/2023	3:27:08 PM	38154	9	244.1	0.1	10.9111	6.562	49.9	0.45	0	0	0.0314	0	0	0
8/23/2023	3:27:09 PM	39220	10	244.06	0.1	10.94	6.562	49.9	0.45	0.049	0	0.049	0	0	0
8/23/2023	3:27:10 PM	40227	10	243.97	0.1	10.97	6.562	49.93	0.45	0.0458	0	0.0458	0	0.0458	0
8/23/2023	3:27:11 PM	41294	10	243.89	0.1	10.91	6.562	49.9	0.45	0.03	0	0.03	0	0	0
8/23/2023	3:27:12 PM	42300	10	243.8	0.1	11	6.562	49.9	0.45	0	0	0	0	0	0
8/23/2023	3:27:13 PM	43367	10	243.8	0.1	11	6.562	49.9	0.45	0	0	0	0	0	0
8/23/2023	3:27:14 PM	44314	9	243.8555	0.1	10.9444	6.562	49.9667	0.45	0.0497	0	0.0497	0	0.0471	0
8/23/2023	3:27:15 PM	45381	10	243.82	0.1	10.9	6.562	49.9	0.45	0.098	0	0	0	0	0
8/23/2023	3:27:16 PM	46387	10	243.7	0.1	10.9	6.562	49.93	0.45	0	0	0	0	0.0458	0
8/23/2023	3:27:17 PM	47454	10	243.8	0.1	10.9	6.562	49.96	0.45	0	0	0	0	0.049	0

Data Logging to Excel Sheet



Software Development



```
File Edit Selection View Go Run Terminal Help
NIALM_SD.ipynb X
C: > Users > weiha > OneDrive > Desktop > TEST > NIALM_SD.ipynb > M*NIALM SD MACHINE LEARNING > import pandas as pd
+ Code + Markdown + Run All + Restart + Clear All Outputs + Variables + Outline ...

NIALM SD MACHINE LEARNING

# Import required libraries
import pandas as pd
import numpy as np
import datetime as dt
import sklearn
from scipy import stats
import matplotlib.pyplot as plt

from sklearn.neural_network import MLPClassifier
from sklearn.neural_network import MLPRegressor
from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import StandardScaler

# Import necessary modules
from sklearn import preprocessing
from sklearn.preprocessing import MaxAbsScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error
from math import sqrt
from sklearn.metrics import accuracy_score

[31] ✓ 0.0s

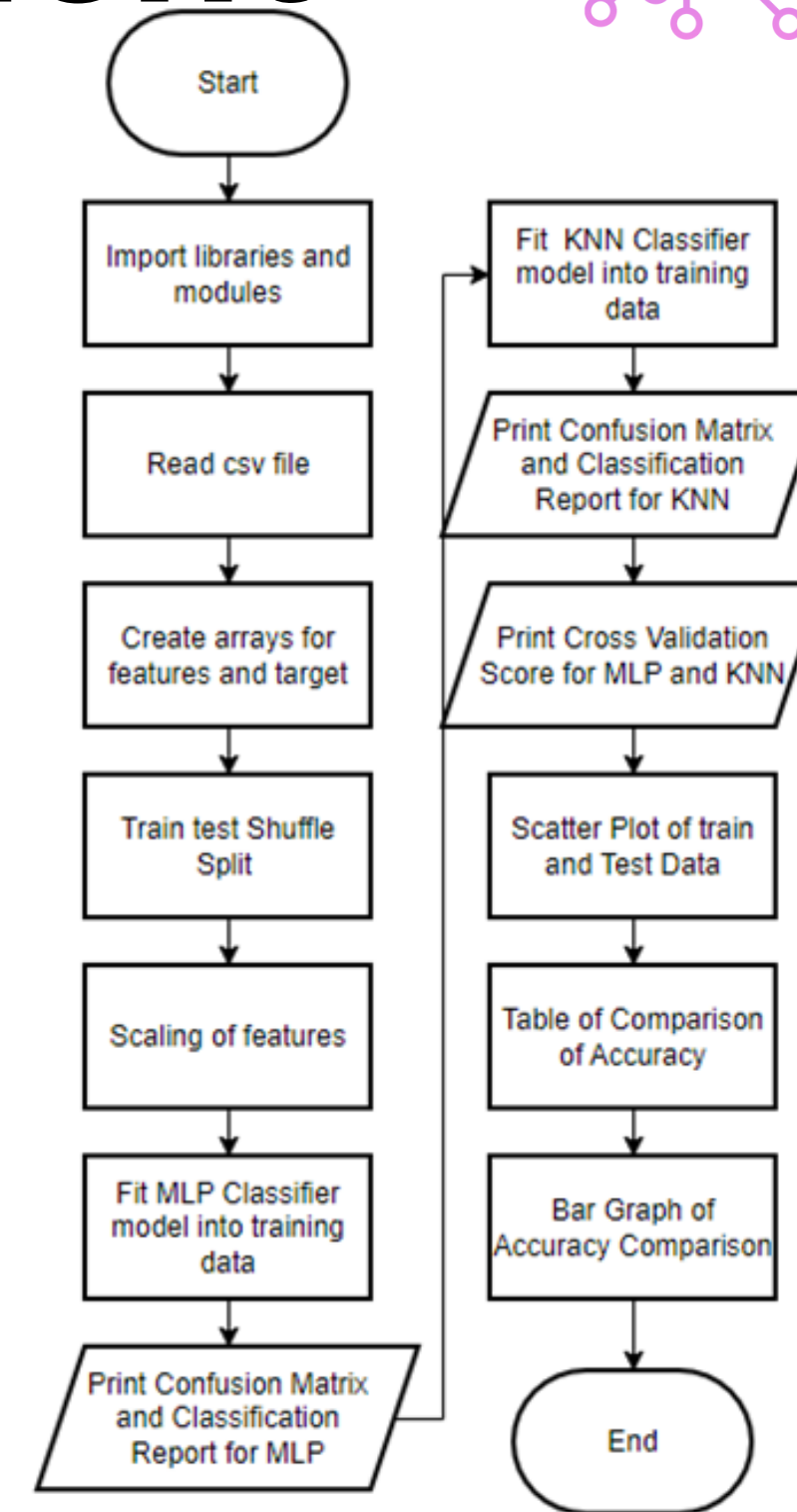
Read csv File

# Counts the number of columns in the file
with open('TEST_ML.csv') as x:
    ncols = len(x.readline().split(','))

# Reads the csv file, skipping the first 4 columns
df = pd.read_csv('TEST_ML.csv', usecols=range(4, ncols))

# Prints the shape of the dataframe
print(df.shape)

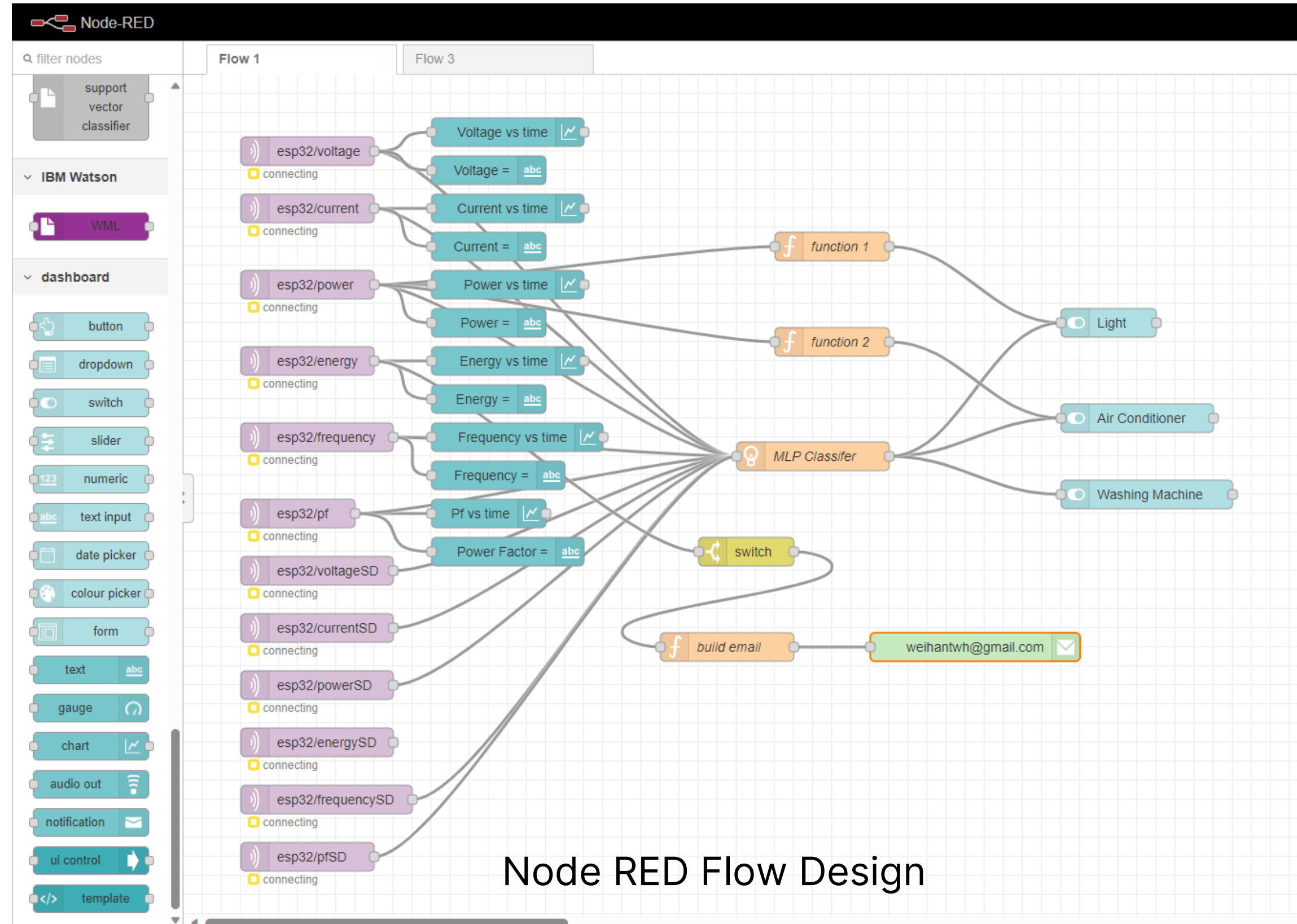
# Prints the statistics of the dataframe
#transpose() is used to make the output more readable
df.transpose()
```



Flowchart for Training ML Models



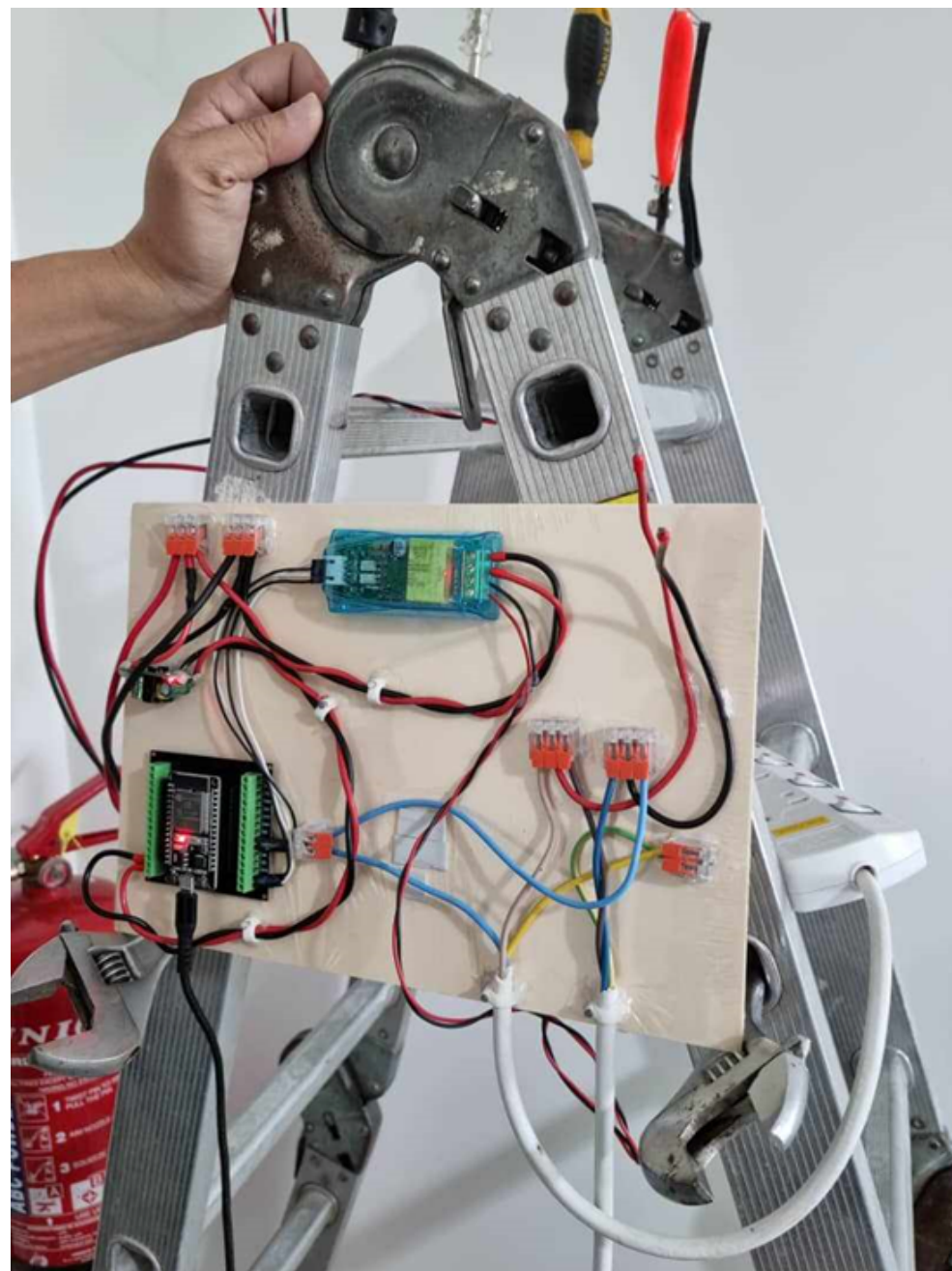
Software Development



Node RED Flow Design

Deployment of Final Prototype

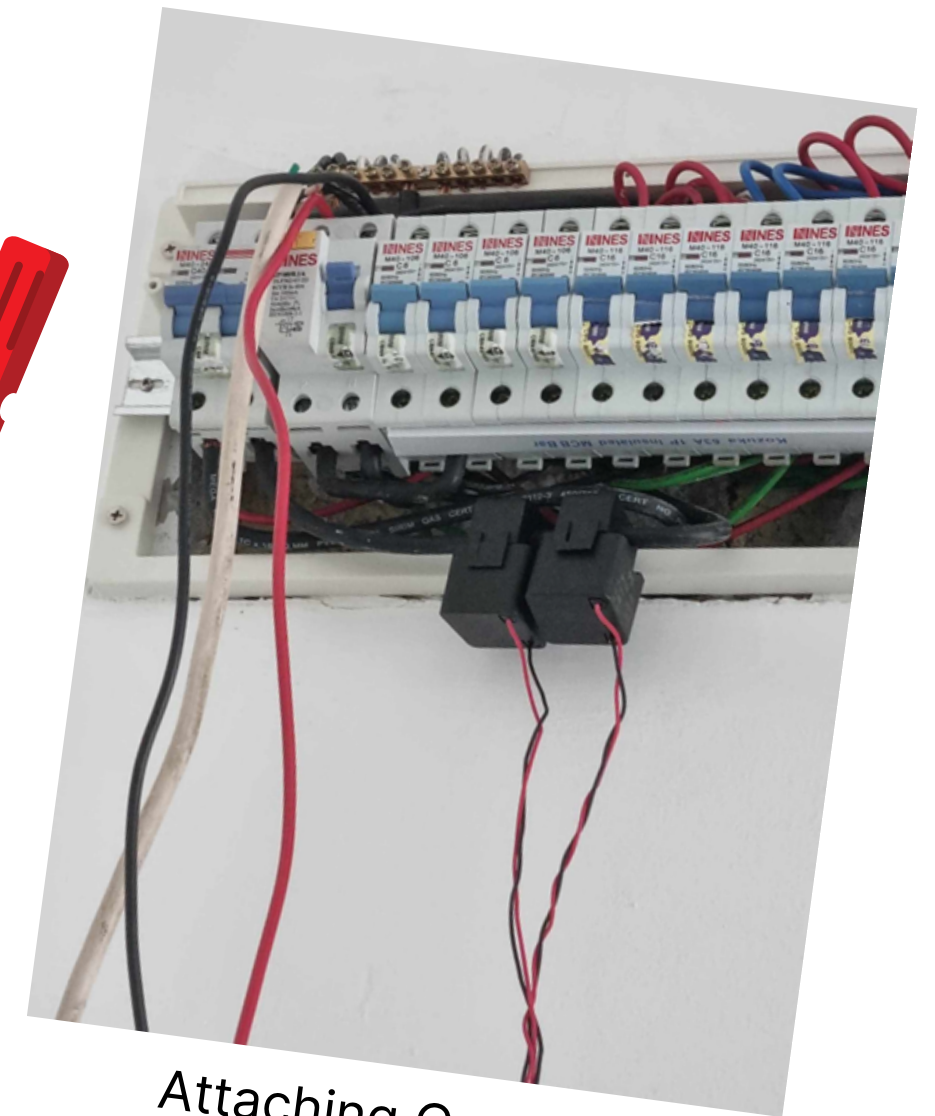
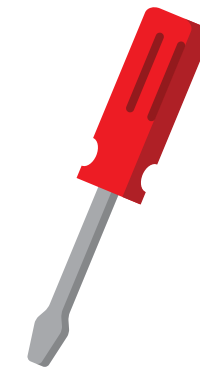
As of 23 August 2023



Final Prototype



Tapping LN wires into MCB



Attaching Current Transformer



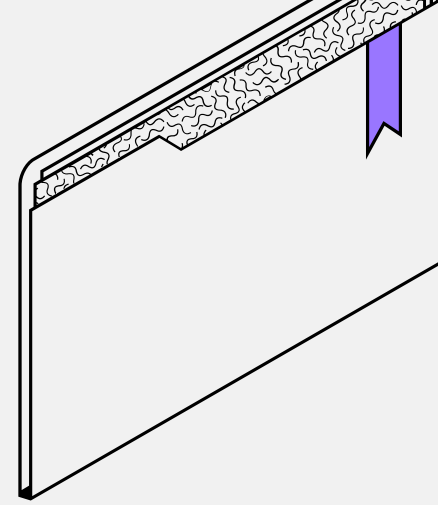
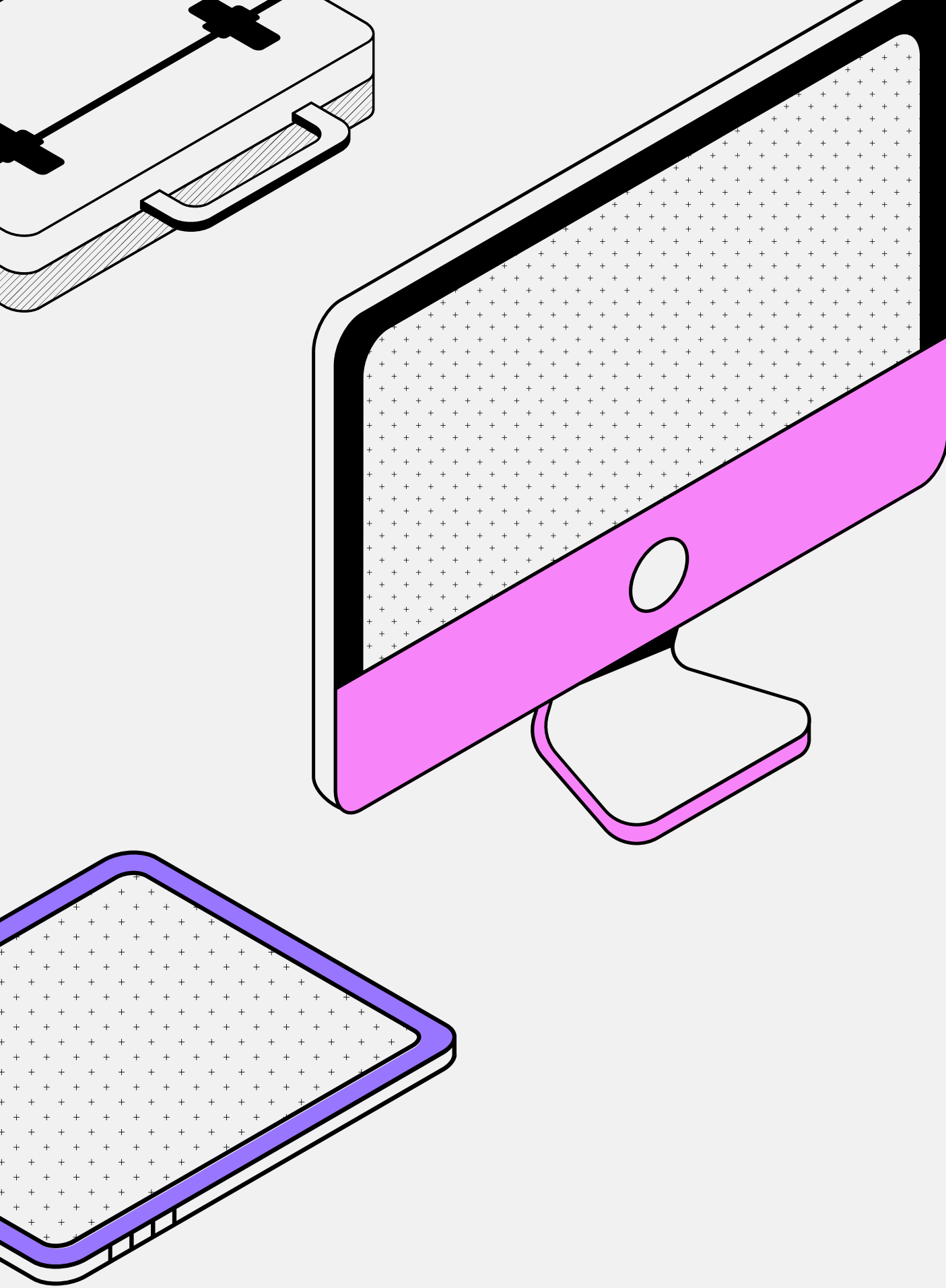
Light



Air Conditioner



Washing Machine



Results and Discussion

MLP



	PREDICTED																
		S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15
A C T U A L	S0	66	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	S1	0	67	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	S2	0	0	76	0	0	0	0	0	0	0	0	0	0	0	0	0
	S3	0	0	0	52	0	0	0	0	0	0	0	0	0	0	0	0
	S4	0	0	0	0	51	0	0	0	0	0	0	0	0	0	0	0
	S5	0	0	0	0	0	66	0	1	1	0	0	0	0	0	0	0
	S6	0	0	0	0	0	0	51	0	0	0	1	0	2	0	0	0
	S7	0	2	0	0	0	0	0	60	0	2	0	0	0	0	0	0
	S8	0	0	0	0	0	0	0	0	55	0	0	0	0	0	0	0
	S9	0	0	0	0	0	0	0	7	4	48	0	0	0	0	0	0
	S10	0	0	0	0	0	0	0	0	0	0	65	0	0	0	0	0
	S11	0	0	0	0	0	0	0	0	0	0	0	64	1	0	0	0
	S12	0	0	0	0	0	0	0	0	0	0	1	0	49	1	0	0
	S13	0	0	0	0	0	0	0	0	0	0	0	0	1	51	0	1
	S14	0	0	0	0	0	0	0	0	0	0	0	0	1	0	62	0
	S15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	51

Confusion Matrix for MLP model

Accuracy on Train Dataset = 98.3%

Accuracy on Test Dataset = 97.29%

KNN



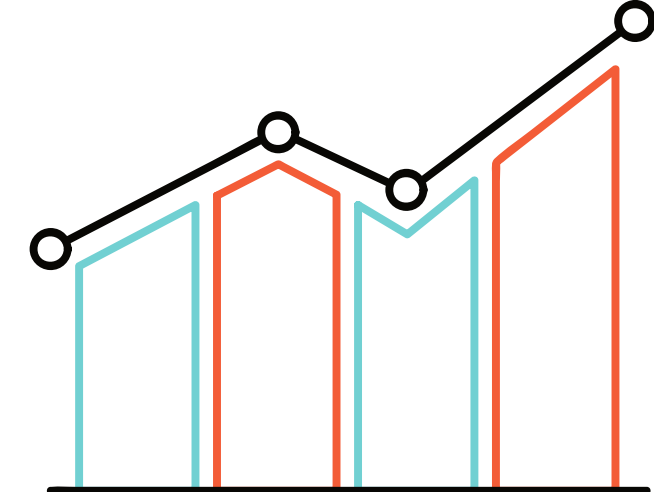
	PREDICTED																
		S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15
A C T U A L	S0	66	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	S1	0	62	5	0	0	0	0	0	0	0	0	0	0	0	0	0
	S2	0	4	60	0	0	0	8	0	0	0	0	1	0	1	0	2
	S3	0	0	0	51	0	1	0	0	0	0	0	0	0	0	0	0
	S4	0	0	0	0	51	0	0	0	0	0	0	0	0	0	0	0
	S5	0	1	0	1	0	57	0	1	0	8	0	0	0	0	0	0
	S6	0	0	1	0	0	0	50	0	0	0	0	1	2	0	0	0
	S7	0	7	0	0	0	0	0	53	1	3	0	0	0	0	0	0
	S8	0	0	0	0	0	0	0	0	55	0	0	0	0	0	0	0
	S9	0	0	0	0	0	8	0	5	4	42	0	0	0	0	0	0
	S10	0	0	1	0	0	0	0	0	0	0	64	0	0	0	0	0
	S11	0	0	0	0	0	0	2	0	0	0	0	62	0	0	1	0
	S12	0	0	0	0	0	0	0	0	0	0	0	1	45	0	5	0
	S13	0	0	0	0	0	0	0	0	0	0	0	0	1	51	0	1
	S14	0	0	0	0	0	0	0	0	0	0	0	0	0	0	63	0
	S15	0	0	0	0	0	0	0	0	0	0	0	0	4	0	2	45

Confusion Matrix for KNN model

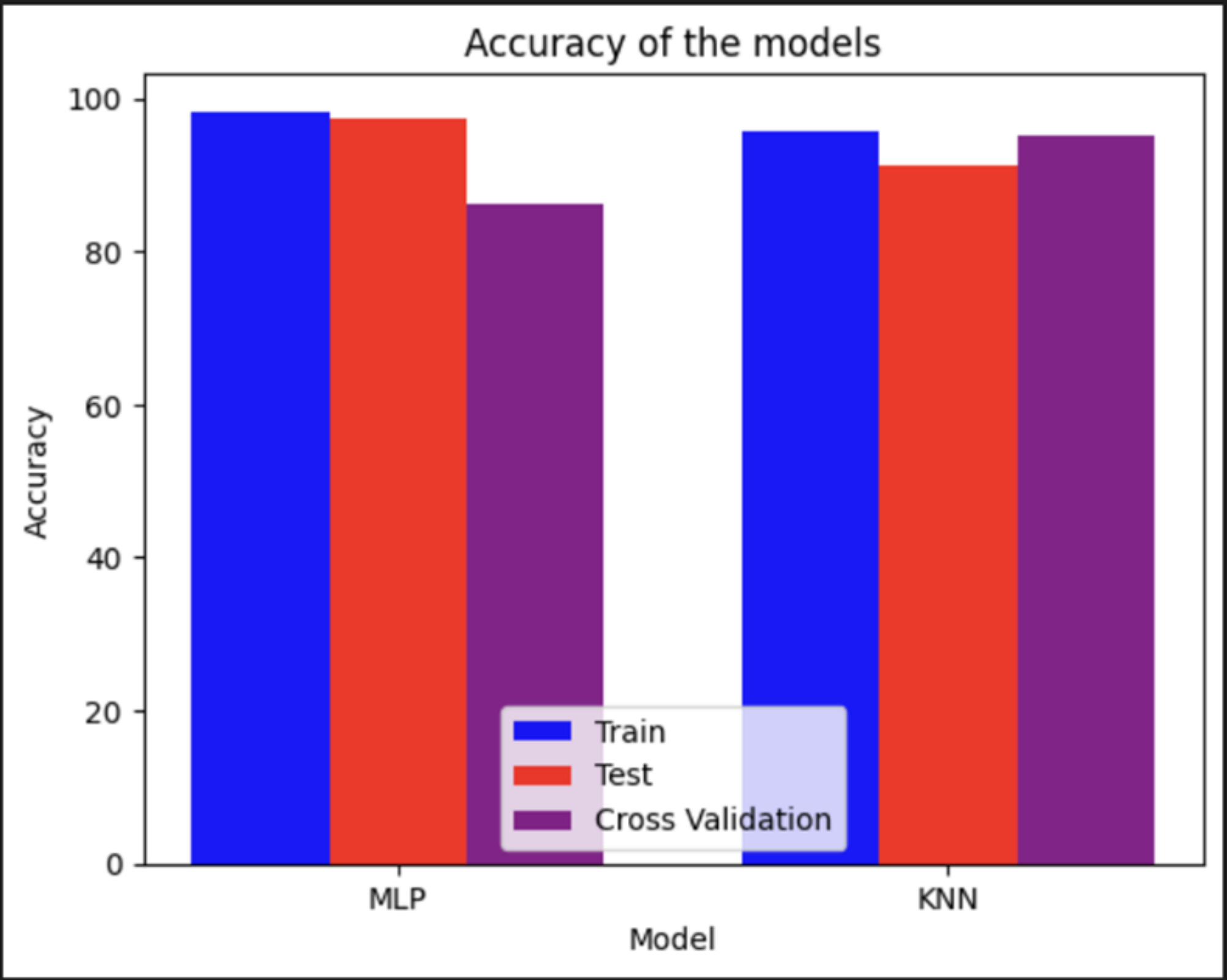
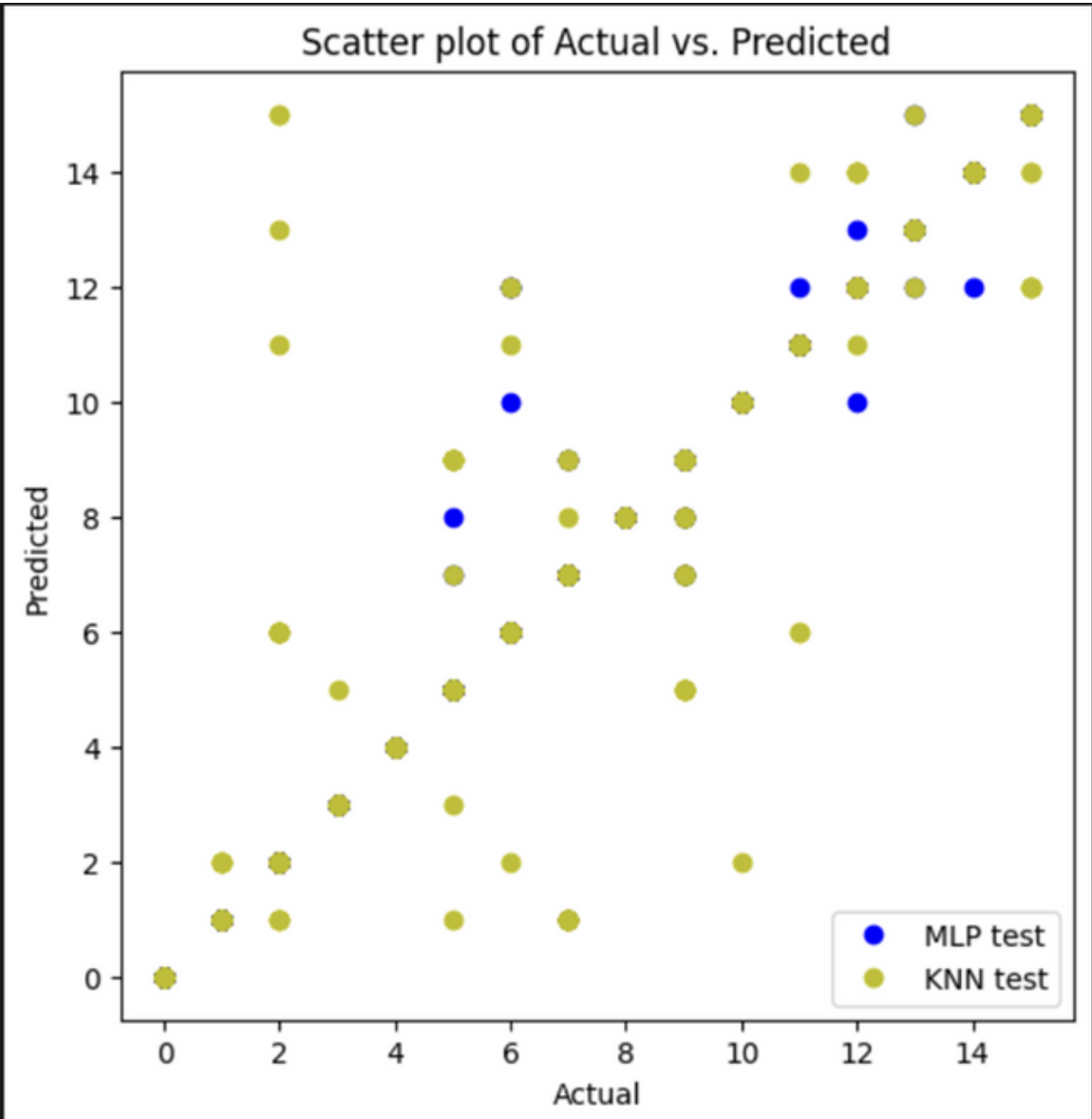
Accuracy on Train Dataset = 95.8%

Accuracy on Test Dataset = 91.35%

Comparison of Accuracy

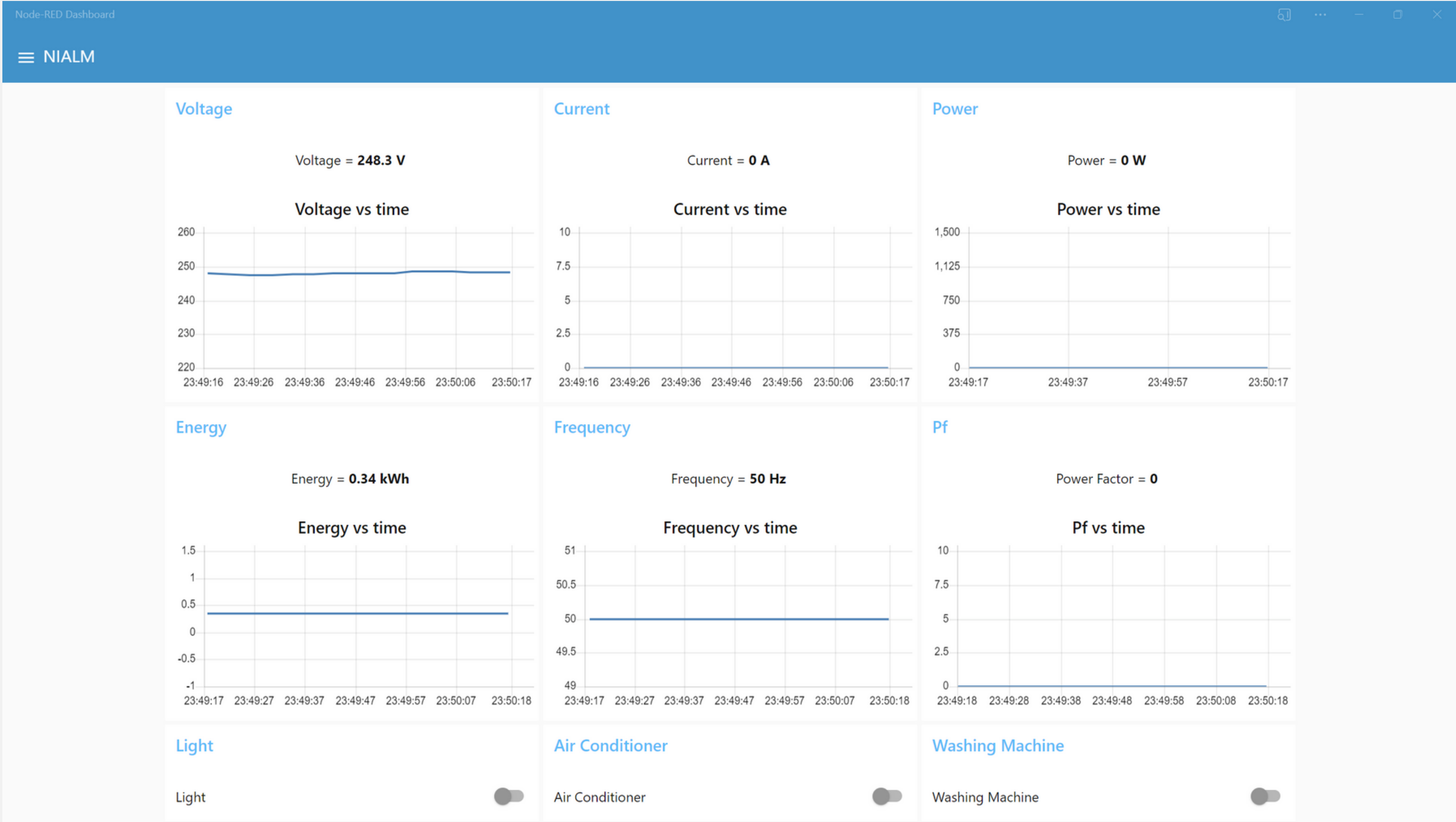


Model	Train(%)	Test(%)	Cross Validation(%)
MLP	98.303571	97.291667	86.303836
KNN	95.803571	91.354167	95.228960



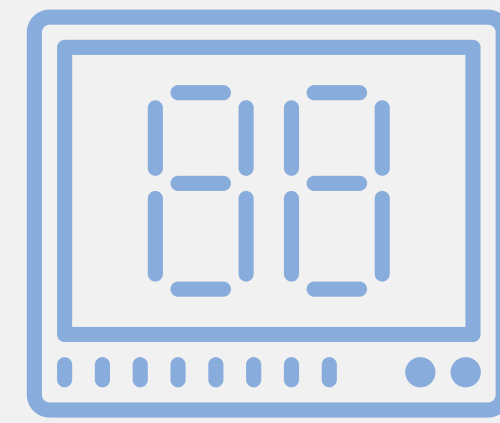
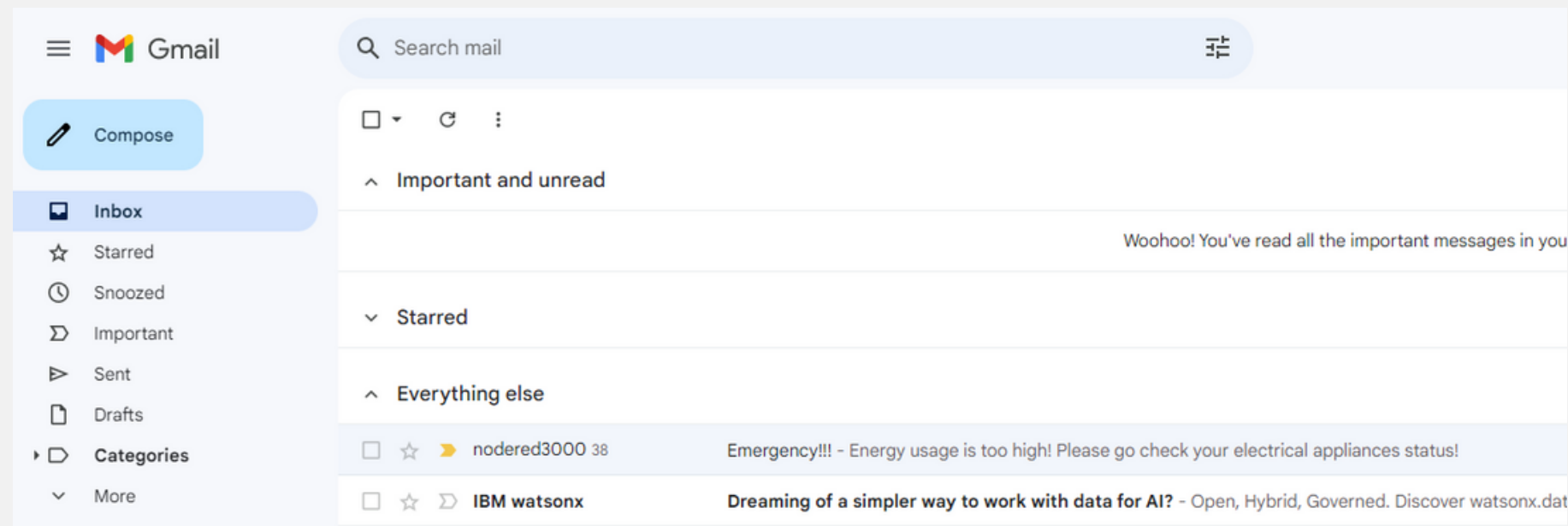
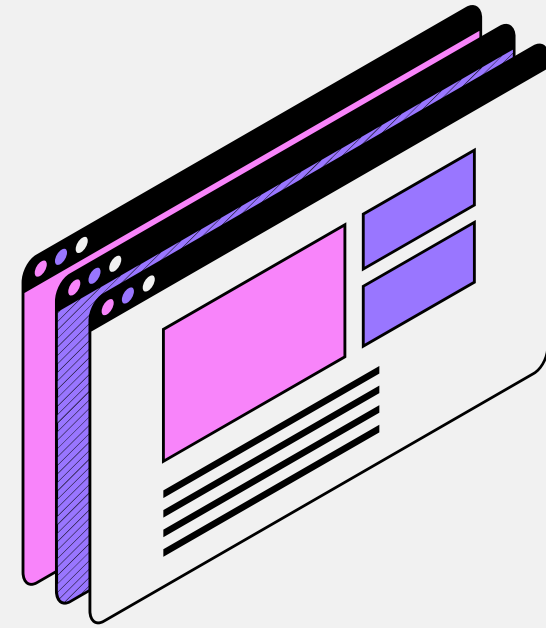
Implementation of IoT

For Real-Time Monitoring and Predictions



Alert Emails

When energy usage exceeds a preset threshold



Liquid Crystal Display

To display energy usage directly



Conclusion

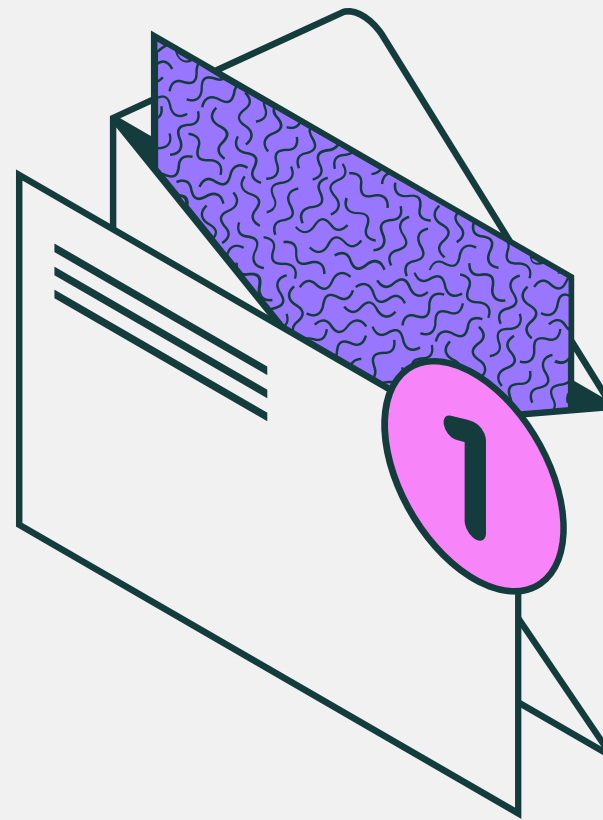
- Successfully upgraded the hardware circuit design.
- Developed the software for NIALM system with ML.
- Implemented IoT into this project.

Recommendations for Future Improvements

- Use a higher quality sensor with higher sampling rate
- Change to a wireless data logging method



Do you have any questions?



The End

